Functional Testing for E-Commerce Web Application

1. Introduction

This project's main aim was to verify that important user functions work as expected and to demonstrate the use of Selenium WebDriver with TestNG and Cucumber.

The application was hosted locally using XAMPP/Apache and tested using Google Chrome. The automation was developed in Eclipse using Java and Maven.

2. Objectives

- To perform functional testing of an E-Commerce web application.
- To automate web application functions using Selenium WebDriver.
- To use different Selenium locators such as ID, XPath, CSS Selector, Class Name and Link Text.
- To use TestNG for test execution and reporting.
- To implement a Cucumber BDD scenario using feature files and step definitions.
- To generate and review TestNG and Cucumber test reports.

3. Tools and Technologies

The following tools and technologies were used during the project:

- Java 17
- Eclipse IDE
- Maven
- Selenium WebDriver
- Google Chrome
- TestNG
- Cucumber
- JUnit
- XAMPP/Apache
- Git/GitHub

4. Application Under Test

The application tested was an E-Commerce web application containing product, contact, shopping cart and checkout functionality. The application was hosted locally and accessed using:

http://localhost/ecomm

5. TestNG Automation

A reusable BaseTest class was created to initialise the browser before the tests. The setup opens Chrome, maximises the browser, deletes cookies, sets a page-load timeout and opens the E-Commerce application.

The HomePageTest was used to test navigation from the home page to the Contact page. The Pages menu was located and a mouse hover was performed using Selenium Actions before selecting Contact.

The CheckoutPageTest was used to test the checkout page. The test entered the customer's first name, last name and company name, selected the United Kingdom from the country dropdown, selected the terms and conditions checkbox and clicked the Place Order button.

Different Selenium locators were used throughout the tests, including Class Name, ID, XPath, CSS Selector and Link Text.

6. TestNG Execution and Result

A testng.xml suite was created containing the HomePageTest and CheckoutPageTest classes. The Testng suite was executed successfully.

Final Testng result:

Total tests run: 2

Passed: 2

Failures: 0

Skipped: 0

7. Cucumber Implementation

Cucumber was used to implement Behaviour Driven Development (BDD). An ECommerce.feature file was created to describe the expected behaviour in a simple Given, When, And and Then format.

The scenario used was:

Add a product to cart and proceed to checkout

- Given I open the e-commerce website
- When I add a product to the cart
- And I proceed to the checkout page

- Then I should see the checkout page

8. Cucumber Execution and Result

A JUnit-based Cucumber TestRunner was created to execute the feature file and generate an HTML report. The scenario was successfully executed.

Scenarios: 1

Passed: 1

Errors: 0

Failures: 0

9. Challenges and Solutions

During testing, the Add to Cart button could not initially be clicked because it was displayed as part of the product's hover content. This was resolved by using Selenium Actions to hover over the product before clicking the button.

The country dropdown also used a custom HTML implementation. The relevant element was identified using a CSS selector and Selenium Actions was used to select the United Kingdom option.

During the initial Cucumber execution, additional Jackson dependencies were required.

10. Conclusion

The Functional Testing for E-Commerce Web Application project was successfully completed using Selenium WebDriver, TestNG and Cucumber. The project demonstrated browser automation, different locator strategies, mouse actions, form and checkbox interaction, BDD implementation and automated reporting.